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Integration Techniques

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To the Student

The purpose of this module is to guide you as you learn integration techniques, to understand how they work and how to use them. You will be reading explanations as well as working through exercises.

As you read, if you are not completely understanding the mathematics, stop and read it again. It might help to read out loud yourself, or have someone else read out loud while you listen. Write down the calculations as you read. Make sure you understand each step and each statement before you go on. If you are not sure about a particular point, discuss it with your classmates or ask your instructor.

As you work exercises, it is likely that you will struggle or become stuck at some point. Consider the poem “Our Real Work” by Wendell Berry:

It may be that when we no longer know what to do
we have come to our real work,
and that when we no longer know which way to go
we have come to our real journey.
The mind that is not baffled is not employed.
The impeded stream is the one that sings.

Allow yourself to struggle. When you are stuck, try something else. Reread or rework the material leading up to where you got stuck. Make sure you do not have a previous mistake that is leading to your stuck-ness. Try something different. Discuss with your classmates an alternative approach. Whatever you do, recognize that your real work is in these moments when you are struggling or stuck. Practice persistence. Ask questions. Keep trying until you get it figured out - the process is as important as reaching a correct solution.

When you finish with a section of this module, you are not done. You then need to practice the techniques you have learned by working through a bunch of homework problems.

So, read carefully, persist when you are struggling, ask questions, practice, and enjoy!

Chapter 1

Integration Techniques

1.1 Review of the Basics

(Basic Integration Rules.)

$\int c \, dx = cx + C$	$\int \sin x \, dx = -\cos x + C$
$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$	$\int \cos x \, dx = \sin x + C$
$\int x^{-1} \, dx = \ln x + C$	$\int \sec^2 x \, dx = \tan x + C$
$\int e^x \, dx = e^x + C$	$\int \csc^2 x \, dx = -\cot x + C$
$\int \frac{dx}{1+x^2} = \arctan x + C$	$\int \sec x \tan x \, dx = \sec x + C$
$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C$	$\int \csc x \cot x \, dx = -\csc x + C$

You should know the integration rules in Key Idea 1.1. If there are any of the rules that you are not comfortable with, please ask! If there are any basic rules that are missing from the list, add them in. Additionally, you are expected to know basic substitution. Exercise 1.1 will help you review these integration rules, as well as some useful algebraic techniques.

Calculate each integral.

a. $\int \frac{1}{2-3x} \, dx$

b. $\int \sec^2 x \, dx$

c. $\int \tan \theta d\theta$

d. $\int e^{\sin x} \cos x dx$

e. $\int (1 + \tan^2 x) dx$

f. $\int \frac{x^2 - 2x - 3}{x + 1} dx$

g. $\int \frac{\ln x}{x} dx$

h. $\int \sec x \tan x dx$

i. $\int \frac{x}{\sqrt{1 + 4x^2}} dx$

j. $\int \frac{e^t - e^{-t}}{e^t + e^{-t}} dt$

k. $\int \frac{1}{e^x + e^{-x}} dx$

Hint: first multiply by $1 = \frac{e^x}{e^x}$, then use a substitution.

l. $\int \frac{1}{\sqrt{9 - x^2}} dx$

Hint: first use $\sqrt{9 - x^2} = 3\sqrt{1 - (x/3)^2}$, then use a substitution.

m. $\int \frac{3x + 1}{\sqrt{1 - x^2}} dx$

Hint: first split up the fraction using the rule that $\frac{a+b}{c} = \frac{a}{c} + \frac{b}{c}$.

n. $\int \frac{x + 1}{-x^2 + 3x + 4} dx$

Hint: first reduce the rational function (factor the denominator).

o. $\int \frac{x^2 + 2x - 1}{x + 4} dx$

Hint: first use long division to simplify the improper rational function.

p. $\int \frac{1}{x^2 + 8x + 17} dx$

Hint: first complete the square, then use a substitution.

1.2 Integration by Parts

First, let's see how we can come up with an integration formula using the product rule for differentiation. As you read through, make sure you understand each step. Start with the product rule for differentiation of uv , and write it in differential notation:

$$d(uv) = u dv + v du.$$

Integrating both sides of the product rule gives

$$\int d(uv) = \int (u dv + v du) \quad (1.1)$$

$$uv = \int u dv + \int v du \quad (1.2)$$

Question: What rule or theorem allows us to simplify from equation 1.1 to equation 1.2?

Solving equation 1.2 for $\int u dv$ gives

$$\int u dv = uv - \int v du$$

which is the rule for integration by parts.

(Integration by Parts.)

$$\int u dv = uv - \int v du$$

Integrating by parts requires a number of steps, and sometimes trial and error. We will go through the steps to integrate $\int x \sin x dx$.

Step 1: When using the integration by parts formula, we first need to choose u and dv so that our integral looks like $\int u dv$. For the integral in the table below, make two different choices for u and dv . *Note:* Initially, don't worry about whether the choices are good or not as long as $u dv = x \sin x dx$. For now, you need only choose u and dv ; we will fill in the other columns in the tables in later steps. I've made the first choice for you.

$\int x \sin x \, dx$	$u, \, dv$	$du, \, v$	Integration by Parts
First Choice	$u = \sin x$ $dv = x \, dx$	$du = \cos x \, dx$ $v = \frac{1}{2}x^2$	$\int x \sin x \, dx =$ $\frac{1}{2}x^2 \sin x - \int \frac{1}{2}x^2 \cos x \, dx$
Second Choice			

Step 2: For each choice in the table, calculate du and v by differentiating u and anti-differentiating/integrating dv . Fill in the appropriate column with your results. Again, I've done the first ones for you.

Step 3: Rewrite the integral using the Integration by Parts formula and fill in the final column, like I did for the First Choice.

Step 4: Now we need to analyze the new integral. Mark a choice as GOOD if the new integral is something we can integrate or is nicer than the one we started with. Mark a choice MAYBE if it is at least no worse than the one we started with. Otherwise mark the choice as BAD.

Step 5: Using the GOOD choice, finish calculating $\int x \sin x \, dx$.

We will use the same process to evaluate $\int x^2 e^x \, dx$. Start by filling in the table and analyzing each choice as GOOD, MAYBE, or BAD.

$\int x^2 e^x \, dx$	$u, \, dv$	$du, \, v$	Integration by Parts
First Choice			
Second Choice			

Now, use the GOOD choice to finish calculating $\int x^2 e^x \, dx$. You will have to use integration by parts a second time.

For $\int \ln x \, dx$, we cannot let $dv = \ln x \, dx$ since we don't know the antiderivative of $\ln x$. So we must let $u = \ln x$. Determine dv , du and v and apply integration by parts to calculate $\int \ln x \, dx$.

Now calculate $\int e^x \sin(3x) \, dx$. Start by using the same GOOD or MAYBE choices and integration by parts multiple times.

1.3 Trigonometric Integrals

(Half-Angle and Double-Angle Identities.)

Half-Angle Identities	Double-Angle Identities
$\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$	$\sin 2\theta = 2 \sin \theta \cos \theta$
$\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$	$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ $= 2 \cos^2 \theta - 1$ $= 1 - 2 \sin^2 \theta$

Use Half-Angle Identities (see Key Idea 1.3) to calculate $\int \sin^2 x dx$ and $\int \cos^2 x dx$.

Your answers to Exercise 1.3 probably involve the angle $2x$. Use Double-Angle Identities to rewrite your answers to involve only the angle x . This may be useful later when we are doing trig substitutions in Section 1.4.

Use Half-Angle Identities to calculate $\int \cos^2 x \sin^2 x dx$.

(Substitutions and the Pythagorean Identities.)

Substitution	Pythagorean Identity
If $u = \sin \theta$, then $du =$.	$\cos^2 \theta + \sin^2 \theta = 1$
If $u = \cos \theta$, then $du =$.	$\cos^2 \theta =$ $\sin^2 \theta =$

Fill in the the du parts of the two substitutions in Key Idea 1.3. Fill in the missing parts of the alternate versions of the Pythagorean Identity in Key Idea 1.3 by rewriting $\cos^2 \theta + \sin^2 \theta = 1$ in appropriate ways.

Calculate $\int \sin^2 x \cos x dx$ using an appropriate substitution from Key Idea 1.3.

Now calculate $\int \sin^3 x \cos^2 x dx$ in a similar way after first using the Pythagorean Identity to rewrite $\sin^2 x$ in terms of $\cos^2 x$.

Look back over all of the exercises with integrals of the form $\int \sin^m x \cos^n x dx$. The approach you take to integrating depends on whether m and n are odd or even. Fill in the missing words in the rules below:

- For $\int \sin^m x \cos^n x dx$, if m and n are _____ then use Half-Angle Identities.
- For $\int \sin^m x \cos^n x dx$, if m or n is _____ then use the Pythagorean Identity and Substitution with $u = \cos x$ if _____ and $u = \sin x$ if _____.

(Tan & Sec Identities and Integrals.)

Identity	Integral
$1 + \tan^2 x = \sec^2 x$	$\int \tan x dx = \ln \sec x + C$ $\int \sec x dx = \ln \sec x + \tan x + C$

The only new information in Key Idea 3 may be $\int \sec x dx = \ln |\sec x + \tan x| + C$. The calculation begins as follows:

$$\begin{aligned} \int \sec x dx &= \int \sec x \left(\frac{\sec x + \tan x}{\sec x + \tan x} \right) dx \\ &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx \end{aligned}$$

Finish the calculation using a u -substitution with $u = \sec x + \tan x$.

We want to use some combination of Key Idea 1.3, substitution, and integration by parts to calculate integrals of the form $\int \tan^m x \sec^n x dx$. Whenever possible, use the identity $1 + \tan^2 x = \sec^2 x$ and do a substitution with $u = \sec x$ or $u = \tan x$. For the substitution, you'll need to calculate du . If $u = \sec x$ then $du =$ _____. If $u = \tan x$ then $du =$ _____. When using one of these substitutions, you must 'save' out the du part and then rewrite the rest of the integral in terms of u using the identity.

Calculate each integral.

a. $\int \tan^3 x \sec^5 x dx$

b. $\int \tan^4 x \sec^4 x dx$

Look back over the integrals of the form $\int \tan^m x \sec^n x dx$. The approach you take depends on whether m and n are odd or even. Fill in the missing words in the rules below:

- For $\int \tan^m x \sec^n x dx$, if m is _____ and $n \geq 1$ then use $u = \sec x$.
- For $\int \tan^m x \sec^n x dx$, if n is _____ then use $u = \tan x$.

There are no clear rules for other integrals of the form $\int \tan^m x \sec^n x dx$, so you have to just try stuff like substitution, integration by parts, and Key Idea 1.3. Calculate each integral. Be persistent!

a. $\int \sec^3 x dx$

b. $\int \tan^4 x dx$

1.4 Trigonometric Substitution

(Useful Identities.)

Pythagorean Identities	Trig Identities	Trig Identities
$1 - \sin^2 \theta = \cos^2 \theta$	$\sin \theta = \frac{a}{c} = \frac{\text{opp}}{\text{hyp}}$	$\csc \theta = \boxed{}$
$\sec^2 \theta - 1 = \tan^2 \theta$	$\sec \theta = \frac{c}{b} = \frac{\text{hyp}}{\text{adj}}$	$\cos \theta = \boxed{}$
$\tan^2 \theta + 1 = \sec^2 \theta$	$\tan \theta = \frac{a}{b} = \frac{\text{opp}}{\text{adj}}$	$\cot \theta = \boxed{}$

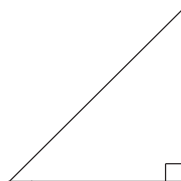


Figure 1.1: Right triangle

On the triangle in Figure 1.1, label the lower left angle θ , the vertical leg a , the horizontal leg b , and the hypotenuse c . In Key Idea 1.4, write down the missing trig identities for $\csc \theta$, $\cos \theta$ and $\cot \theta$.

You should *know* the Pythagorean Identity $\sin^2 \theta + \cos^2 \theta = 1$. What do you get if you divide both sides by $\cos^2 \theta$ and simplify? What if you divide by $\sin^2 \theta$?

We will now use the identities and a “trig substitution” to integrate $\int \frac{1}{x^2 + 1} dx$. Of course, we know the answer should be $\arctan x + C$, but trig substitution gives us a way to *calculate* it instead of having to just *know* it.

We begin by setting up a substitution equation. There are two methods for doing this; we’ll look at both.

Triangle/Geometry Method: Draw a right triangle like the one in Figure 1.1. Since our integral contains a sum of squares $x^2 + 1 = x^2 + 1^2$, label the vertical leg of the triangle x , the horizontal leg 1 , and the lower left angle θ . Using the Pythagorean Theorem, $a^2 + b^2 = c^2$, your hypotenuse should be $c = \sqrt{x^2 + 1}$. Notice that $c^2 = x^2 + 1$, matching the sum of squares in our integral.

Next, write down what you get when you apply the Trig Identities in Key Idea 1.4 to your triangle. Which one gives the nicest way to write x as a function of θ , i.e., in terms of $\sin \theta$, $\sec \theta$, or $\tan \theta$?

You should get that $\tan \theta = \frac{x}{1} = x$.

This is our substitution equation:

$$\tan \theta = x.$$

Note also that, from the triangle, we have $\sec \theta = \sqrt{x^2 + 1}$, and hence

$$\sec^2 \theta = x^2 + 1.$$

Thus, using the Triangle/Geometry Method, we have the substitution equation $\tan \theta = x$ and the relation $\sec^2 \theta = x^2 + 1$. Next, we will consider a different method to determine the same equations.

Algebra Method: Notice that $x^2 + 1$ is of the form (variable)² + 1, and, considering the Pythagorean Identities in Key Idea 1.4, $\tan^2 \theta + 1$ is of the same form. So we want $x^2 + 1 = \tan^2 \theta + 1$. Solve the equation for x . Did you get the same substitution equation,

$$x = \tan \theta,$$

as with the Triangle/Geometry Method?

Also, knowing that $x^2 + 1 = \tan^2 \theta + 1$, notice that the identity $\tan^2 \theta + 1 = \sec^2 \theta$ can be rewritten as

$$x^2 + 1 = \sec^2 \theta.$$

Thus, using the Algebra Method, we get the same the substitution equation and relation as with the Triangle/Geometry Method.

From either Method, we end up with $x = \tan \theta$ and $x^2 + 1 = \sec^2 \theta$. Before we can perform the substitution, just like with u -substitutions, we need to calculate dx . Using $x = \tan \theta$, calculate $dx = \underline{\hspace{2cm}}$. Substituting, simplifying, and integrating, we get

$$\int \frac{1}{x^2 + 1} dx = \int \frac{1}{\sec^2 \theta} \cdot \sec^2 \theta d\theta = \int 1 d\theta = \theta + C.$$

Finally, as with any substitution method, we need to undo the substitution. Using the transition rule on $x = \tan \theta$, we get $\arctan x = \theta$. Thus,

$$\int \frac{1}{x^2 + 1} dx = \arctan x + C.$$

This is exciting! You no longer need to *memorize* a formula for $\int \frac{1}{x^2 + 1} dx$ because you can *calculate* it instead!

Use similar trig substitutions and whichever method you prefer to calculate each integral. It is always possible to use one of the substitutions $x = \sin \theta$, $x = \sec \theta$, or $x = \tan \theta$, or similar substitutions with constant coefficients (though others may work as well). You might have to play around some to figure out a correct substitution, triangle, or identity.

a. $\int \frac{dx}{(x^2 + 5)^{3/2}}$

b. $\int \frac{dx}{x^2 \sqrt{16 - x^2}}$

c. $\int \frac{1}{x \sqrt{x^2 - 4}} dx$

d. $\int \frac{dx}{\sqrt{4x^2 - 25}}$

e. $\int \frac{x^2}{\sqrt{9 - x^2}} dx$

Let's review what we've just done. What is the pattern for trig substitutions? For each expression in Key Idea 1.4, fill in the trig substitution (e.g. $x = a \tan \theta$) you would use if you see the expression in your integral. Also fill in the appropriate identity or triangle, whichever you prefer to use.

(Trig Substitutions.)

Expression	$a^2 - x^2$	$a^2 + x^2$	$x^2 - a^2$
Trig Substitution			
Identity or Triangle			

Strategy 1: Simplify the right side of Equation 1.5 and collect like terms:

$$\begin{aligned} 4x - 1 &= A(x - 1) + B(x + 2) \\ &= Ax - A + Bx + 2B \\ &= (A + B)x + (-A + 2B). \end{aligned}$$

Set the coefficients of like terms from each side of the resulting equation

$$4x - 1 = (A + B)x + (-A + 2B)$$

equal to each other:

$$\begin{aligned} \text{coefficients of } x: \quad 4 &= A + B \\ \text{constant terms:} \quad -1 &= -A + 2B. \end{aligned}$$

Now, solve this system of equations for A and B .

Strategy 2: Choose convenient values of x in Equation 1.5 to make parts become zero:

$$x = 1: \quad 4 - 1 = A(1 - 1) + B(1 + 2) \text{ becomes } 3 = 3B, \text{ so } B = 1.$$

Now, try this using $x = -2$ to determine the value of A .

Hopefully using both methods you obtained $A = 3$, $B = 1$, and hence the decomposition $\frac{4x - 1}{(x + 2)(x - 1)} = \frac{3}{x + 2} + \frac{1}{x - 1}$.

More generally, the steps for partial fraction decomposition are:

1. Factor the denominator.
2. Determine the fractions in your decomposition, like in Equation 1.4.
3. Solve for values of the unknowns using Strategies 1 and/or 2.
4. Integrate the resulting, simpler, rational functions.

Calculate each integral using partial fraction decomposition.

a. $\int \frac{-2x + 5}{x^2 + x - 2} dx$

b. $\int \frac{2}{2x^2 + x} dx$

In the previous exercises you decomposed into fractions with *linear* denominators, i.e., denominators of the form $ax + b$. Each of the resulting fractions had only a constant in the numerator. In other words, we have the following rule:

For a linear factor of $ax + b$ in the denominator, use $\frac{A}{ax + b}$ in your decomposition.

Note: You may use any letters in place of A , such as B or C , as long as you don't use the same letter twice in your decomposition.

Next, consider the decomposition

$$\frac{3x^2 - x + 3}{x^3 + x} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1}. \quad (1.6)$$

- a. Notice that the numerator of the left-hand fraction is a quadratic polynomial, while the denominator is a reducible cubic. Characterize the numerators and denominators of the right-hand fractions as: constant, linear, reducible quadratic, irreducible quadratic, or cubic.
- b. Factor $x^3 + x$. Do the factors match the denominators of the decomposition fractions in Equation 1.6?
- c. To calculate $\int \frac{3x^2 - x + 3}{x^3 + x} dx$, first decompose $\frac{3x^2 - x + 3}{x^3 + x}$ using Strategies 1 and/or 2 to solve for the A , B , and C in Equation 1.6.
- d. Next, use your decomposition to integrate $\int \frac{3x^2 - x + 3}{x^3 + x} dx$.

In the previous example you decomposed into a fraction with a linear denominator, and another with an *irreducible quadratic* denominator. For the fraction with the irreducible quadratic denominator, you used a *linear* numerator $Bx + C$. This leads to the following rule:

For an irreducible quadratic factor $ax^2 + bx + c$ in the denominator, use

$$\frac{Ax + B}{ax^2 + bx + c}$$

in your decomposition.

Note: The letters don't matter, only that you don't use the same letter twice in your decomposition.

As a third example, consider

$$\frac{5x + 9}{x^3 + 6x^2 + 9x} = \frac{5x + 9}{x(x + 3)^2} = \frac{A}{x} + \frac{B}{x + 3} + \frac{C}{(x + 3)^2} \quad (1.7)$$

- a. Factor $x^3 + 6x^2 + 9x$. Does your answer agree with Equation 1.7?
- b. Calculate the sum of the three decomposition fractions on the right in Equation 1.7 by first finding the *least* common denominator.

Did you get a fraction with denominator $x(x + 3)^2$? If not then find, fix, and make note of your mistake!

- c. In Equation 1.7, notice that the first two decomposition fractions (with A and B) have linear denominators and constant numerators, matching the pattern in Equation 1.3. However, the third decomposition fraction has a *repeated linear* factor in the denominator, but still has a *constant* numerator. The degree of the numerator is determined by the linear factor $(x + 3)$, and not by the fact that it is a repeated factor.
- d. Use the decomposition in Equation 1.7 and Strategy 1 and/or 2 to calculate $\int \frac{5x + 9}{x^3 + 6x^2 + 9x} dx$.

Considering Equation 1.7, fill in the missing numerators in the rule:

For a repeated linear factor $(ax + b)^2$ in the denominator, use $\frac{\boxed{}}{ax + b} + \frac{\boxed{}}{(ax + b)^2}$ in your decomposition.

Can you generalize your rule for a linear factor repeated n times? Fill in the missing numerators.

For a repeated linear factor $(ax + b)^n$ in the denominator, use $\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \frac{\boxed{}}{(ax + b)^3} + \cdots + \frac{\boxed{}}{(ax + b)^n}$ in your decomposition.

Summarizing the three examples of partial fraction decompositions, we get the following general rules.

(Partial Fraction Decomposition Rules.)

1. For a **linear factor** of $ax + b$ in the denominator, use $\frac{A}{ax + b}$ as a term in your decomposition.
2. For an **irreducible quadratic factor** of $ax^2 + bx + c$ in the denominator, use $\frac{Ax + B}{ax^2 + bx + c}$ as a term in your decomposition.
3. For a **repeated linear factor** of $(ax + b)^n$ in the denominator, use $\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \cdots + \frac{A_n}{(ax + b)^n}$ as part of your decomposition.

Note: We could make other rules, but these rules make our lives easiest when we go to integrate! You might think about what other possible decompositions we could use, and how they relate to these rules.

Use partial fraction decomposition to calculate each integral.

a. $\int \frac{2}{x^3 + 4x^2 + 4x} dx$

b. $\int \frac{x^2 + 2x}{x^4 + x^2} dx$

c. $\int \frac{x^3 + 2x^2 - 3x - 18}{x^2 - x - 6} dx$

Hint: first reduce the rational function using long division or synthetic division.

1.6 Integration Practice

Calculate each integral.

a. $\int e^{2y} \cos 5y dy$

b. $\int \frac{1}{\sqrt{9-x^2}} dx$

c. $\int \frac{x}{\sqrt{9-z^2}} dz$

d. $\int \frac{x^2}{\sqrt{9-x^2}} dx$

e. $\int \sin^3 \theta d\theta$

f. $\int \sin^4 \theta d\theta$

g. $\int \frac{1}{3x-1} dx$

h. $\int \frac{y-1}{y^2+2y-3} dy$

i. $\int \frac{-8}{v^2+2v-3} dv$

j. $\int \frac{11-7x}{(x^2+2x-3)(x-1)} dx$

k. $\int \frac{4t^3 + 2t - 2}{t^2(t^2 + 2)} dt$

l. $\int \frac{1}{x\sqrt{x^2 + 3}} dx$

m. $\int w \ln w dw$

n. $\int \frac{\ln x}{x} dx$

o. $\int \frac{\ln x}{x^2} dx$

p. $\int \arcsin \rho d\rho$

q. $\int \sec^3 \phi \tan^3 \phi d\phi$

r. $\int \sec^4 x \tan^3 x dx$

s. $\int \sin^3 x \cos^3 x dx$

t. $\int \frac{1}{9t^2 - 4} dt$